End-to-End AI Development: Learn, Build, and Deploy Intelligent Systems

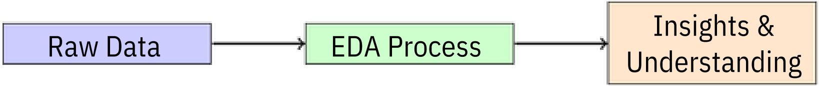
Day 3 : Exploratory Data Analysis (EDA)

Outline

What is Exploratory Data Analysis?

Definition

Exploratory Data Analysis (EDA) is the process of investigating and summarizing datasets to discover patterns, detect anomalies, and check assumptions before applying machine learning mo dels.



Why is EDA Important?

Benefits of EDA:

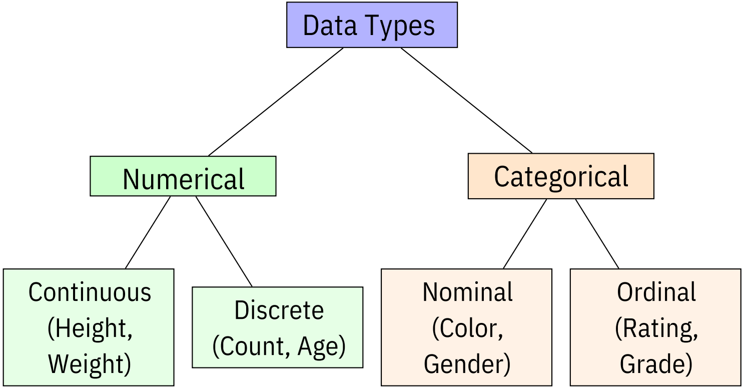
- Understand data structure
- Identify patterns and trends
- Detect outliers and anomalies
- Discover relationships between variables
- Guide feature selection
- Validate data quality

Key Principle

"Garbage In, Garbage Out"

Understanding your data is crucial before building any model!

Types of Data



Numerical vs Categorical Data

Numerical Data

Characteristics:

- Measurable quantities
- Mathematical operations possible
- Examples: temperature, price, distance

Categorical Data

Characteristics:

- Describes qualities or groups
- Limited set of values
- Examples: color, country, yes/no

Why Does This Matter?

Different data types require different analysis techniques and visualizations!

Measures of Central Tendency

Mean
Average value of all data points

Median
Middle value when sorted

Mode
Most frequent value

Example Dataset

Values: 2, 3, 3, 5, 7, 10, 15

- Mean = 6.43 (sum divided by count)
- Median = 5 (middle value)
- Mode = 3 (appears twice)

Measures of Spread (Variability)

How dispersed is your data?

Range

Difference between maximum and minimum values

$$\text{Range} = \text{Max} - \text{Min}$$

Variance

Average of squared deviations from the mean

Standard Deviation

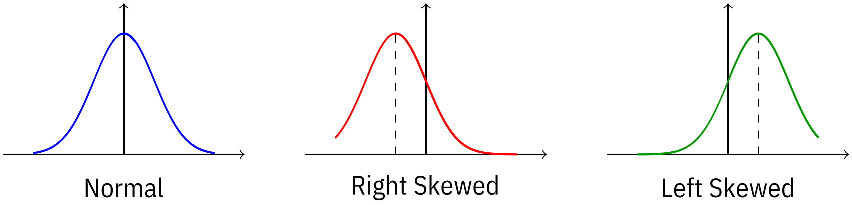
Square root of variance; measures average distance from mean

Interquartile Range (IQR)

Range of the middle 50% of data

$$IQR = Q3 - Q1$$

Distribution Shape



- Normal (Symmetric): Mean = Median = Mode
- Right Skewed: Mean > Median (tail extends right)
- Left Skewed: Mean < Median (tail extends left)

Why Visualize Data?

The Power of Visualization

"A picture is worth a thousand words"

Visualization helps us:

- Quickly identify patterns and trends
- Compare different groups or categories
- Spot outliers and unusual observations
- Communicate findings effectively Make
- data-driven decisions

Rememb er

Choose the right visualization for your data type and question!

Visualizations for Numerical Data

Histogram

- Shows distribution of values
- Displays frequency of ranges (bins)
- Reveals shape and spread

Box Plot

- Shows five-number summary
- Min, Q1, Median, Q3, Max
- Highlights outliers

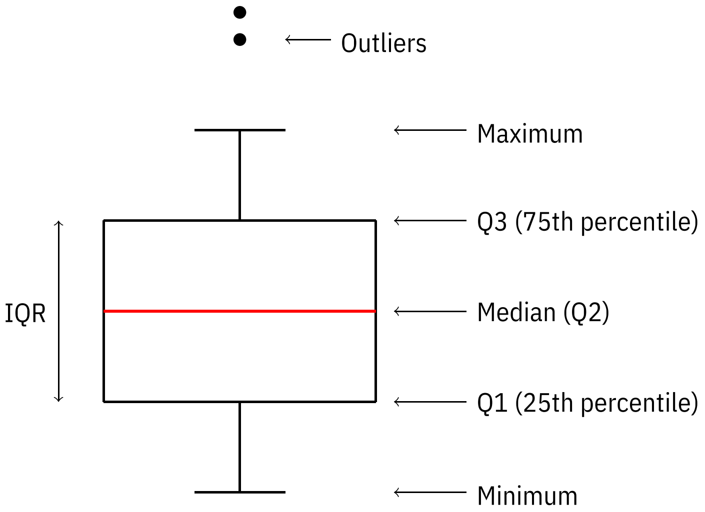
Scatter Plot

- Shows relationship between two variables
- Reveals correlations
- Identifies clusters

Line Plot

- Shows trends over time
- Connects sequential data points
- Good for time series

Understanding Box Plots



Visualizations for Categorical Data

Bar Chart

- Compares categories
- Shows frequencies or counts
- Bars can be horizontal or vertical
- Categories are discrete

When to use:

- Comparing different groups
- Showing counts per category

Pie Chart

- Shows proportions
- Parts of a whole
- Best with few categories (3-6)
- Shows percentages

When to use:

- Displaying percentage breakdown
- Showing composition

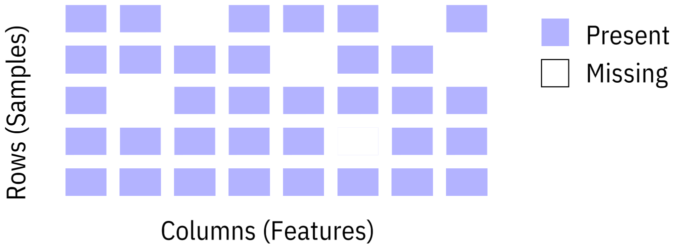
Best Practice

Avoid pie charts with too many categories - use bar charts instead!

Common Data Quality Issues

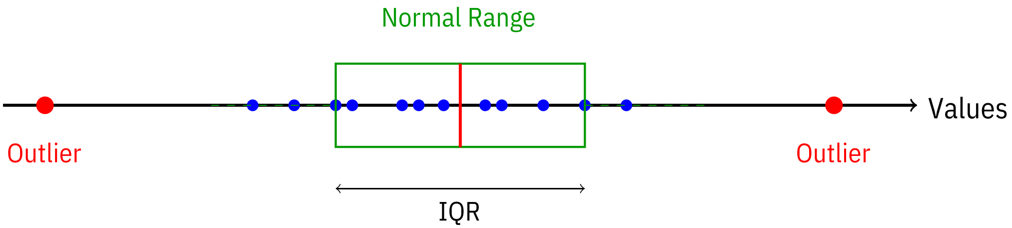
1. Missing Values
   - Incomplete records
   - Empty cells or null values
   - Impact model performance
2. Outliers
   - Extreme values far from others
   - May be errors or genuine rare cases
   - Can skew statistical measures
3. Duplicates
   - Repeated records Can bias
   - analysis Need to be identified and
   - handled

Missing Data Patterns



Key Questions:

- How much data is missing? (percentage)
- Is the missing data random or systematic?
- Which features have the most missing values?

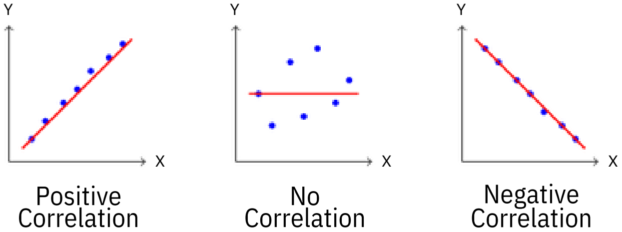


Outlier Detection Methods:

- Visual inspection (box plots, scatter plots)
- Statistical methods (IQR rule: $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$)
- Domain knowledge and context

What is Correlation?

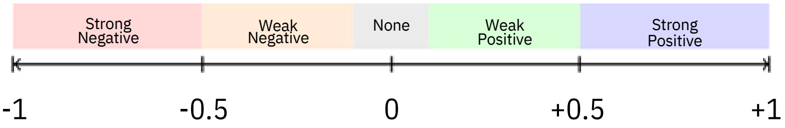
Correlation measures the strength and direction of the linear relationship between two numerical variables.



Correlation Coefficient

Range of Values

Correlation coefficient (r) ranges from -1 to +1



Important Note

Correlation does NOT imply causation! Just because two variables are correlated doesn't mean one causes the other.

Correlation Matrix

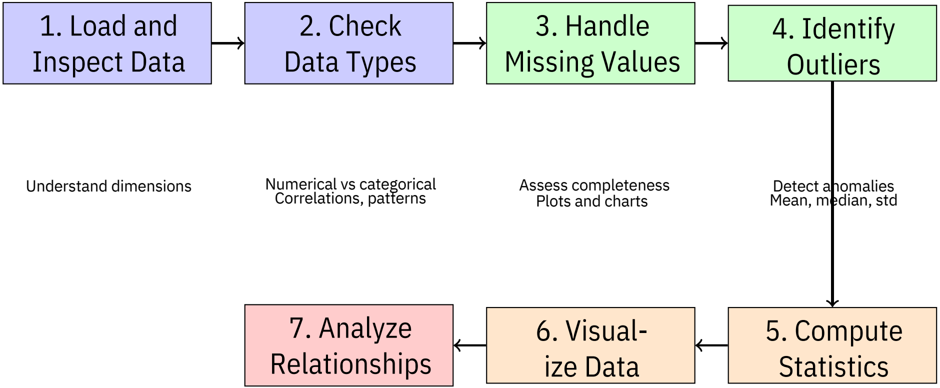
What is it?

- A table showing correlations between all pairs of variables
- Diagonal always equals 1 (variable with itself)
- Symmetric matrix
- Often visualized with a heatmap

Why use it?

- Quickly identify strongly related variables
- Detect multicollinearity (variables that measure similar things)
- Guide feature selection for modeling
- Understand variable relationships at a glance

Step-by-Step EDA Workflow



Key Questions to Ask During EDA

1. About the Data:
   - How many observations and features do I have?
   - What types of variables are present?
   - How much data is missing?
2. About Distributions:
   - What is the typical value for each variable?
   - How spread out is the data?
   - Are there any unusual patterns?
3. About Relationships:
   - Which variables are related to each other?
   - Are there strong correlations?
   - Do I see any obvious patterns or clusters?

EDA Best Practices

Do's:

- √ Start simple, then go deeper
- √ Use multiple visualizations
- √ Document your findings
- √ Question anomalies
- √ Consider domain knowledge
- √ Be systematic and thorough

Don'ts:

- × Skip EDA and jump to modeling
- × Ignore outliers without investigation
- × Rely on a single statistic
- × Forget to check data quality
- × Over-complicate visualizations
- × Assume correlation = causation

Common Pitfalls to Avoid

1. Confirmation Bias

Don’t just look for what you expect to find. Let the data speak!

2. Over-reliance on Automation

Tools are helpful, but understanding what they do is essential.

3. Ignoring Context

Always consider the real-world meaning behind the numbers.

4. Rushing the Process

EDA takes time. Thorough exploration prevents problems later!

Key Takeaways

Rememb er:

1. EDA is the foundation of any machine learning project
2. Understanding your data prevents costly mistakes
3. Different data types require different approaches
4. Visualization is powerful for discovering patterns Always
5. check for data quality issues Correlation helps identify
6. relationships EDA is iterative - you may need multiple
7. passes

”Data is the new oil, but like oil, it needs to be refined before it becomes valuable.”

Next Steps

After completing EDA, you're ready to:

1. Data Preprocessing
   - Handle missing values and outliers
   - Feature engineering
   - Data transformation
2. Feature Selection
   - Choose relevant variables
   - Remove redundant features
3. Model Building
   - Select appropriate algorithms
   - Train and evaluate models

Thank You For your Attention

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